HW 4

1. (a) by HW3 Q1 · (d), $\hat{\lambda}_{MLE} = \bar{X}_n$

the maximum likelihood estimate is 65.

by equivariance / invariance of MLE,

$$\widehat{\frac{\lambda_0}{120}}_{MLE} = \frac{\hat{\lambda}_{MLE}}{120} \approx 54.17\%$$

(b) $$\ell_5(\pi) \propto \left(\prod_{i=1}^{2} \pi^{X_i} (1-\pi)^{100-X_i}\right)\left(\prod_{i=3}^{5} \pi^{X_i} (1-\pi)^{120-X_i}\right)$$

$$\propto \pi^{\sum X_i} (1-\pi)^{560 - \sum X_i}$$

$$\ell_5(\pi) = \text{const} + (\textstyle\sum X_i) \log \pi + (560 - \sum X_i) \log(1-\pi)$$

F.O.C. $$\dot{\ell}_5(\pi) = (\textstyle\sum X_i)\frac{1}{\pi} - (560 - \sum X_i)\frac{1}{1-\pi} = 0$$

S.O.C. $$\ddot{\ell}_5(\pi) = -(\textstyle\sum X_i)\frac{1}{\pi^2} - (560 - \sum X_i)\frac{1}{(1-\pi)^2} < 0$$

then $$\hat{\pi}_{MLE} = \frac{1}{560}\sum_{i=1}^{n} X_i$$

with the data, $\hat{\pi} = 58.04\%$

(c) $$\ell_5(\alpha, \beta) = \prod_{t=1}^{5} e^{-(\alpha+\beta t)} \frac{(\alpha+\beta t)^{X_t}}{X_t!}$$

$$\propto \left(\prod_{t=1}^{5} (\alpha+\beta t)^{X_t}\right) e^{-5(\alpha+3\beta)}$$

$$\ell_5(\alpha, \beta) = \text{const} - 5(\alpha+3\beta) + \sum_{t=1}^{5} X_t \log(\alpha+\beta t)$$

Q2.

(a) $0 < t \le \theta$,

$$P(\hat{\theta} \le t) = P(\max(\{X_i\}) \le t)$$
$$= P\left(\bigcap_{i=1}^{n} \{X_i \le t\}\right)$$
$$= \prod_{i=1}^{n} P(X_i \le t) = \left(\frac{t}{\theta}\right)^n$$

then, the CDF of $\hat{\theta}$ is

$$F_\theta(t) = \left(\frac{t}{\theta}\right)^n \mathbb{1}[t \ge 0) \wedge 1, \quad t \in \mathbb{R}.$$

the pdf of $\hat{\theta}$ is

$$f_\theta(t) = n \left(\frac{t}{\theta}\right)^{n-1} \frac{1}{\theta} \mathbb{1}(0 < t < \theta), \quad t \in \mathbb{R}$$

(b) by HW3. Q2-(b), $\hat{\lambda} = (\bar{Y}_n)^{-1}$

$Y_i \overset{iid}{\sim} Exp(\lambda) = Ga(1, \lambda)$

$$\varphi_{Y_i}(t) = \mathbb{E}[e^{tY_i}] = \int e^{ty} \lambda e^{-\lambda y} dy = \frac{\lambda}{\lambda - t}$$

$$\varphi_{\Sigma Y_i}(t) = \mathbb{E}[e^{t(\Sigma Y_i)}]$$
$$= \prod_{i=1}^{n} \mathbb{E}[e^{tY_i}] = \left(\frac{1}{1 - t/\lambda}\right)^n$$

$$\sum_{i=1}^{n} Y_i \sim Ga(n, \lambda)$$

$$\varphi_{\bar{Y}_n}(t) = \mathbb{E}[e^{t \frac{1}{n} \Sigma Y_i}]$$
$$= \left(\frac{1}{1 - t/(n\lambda)}\right)^n \qquad \frac{1}{n}\sum_{i=1}^{n} Y_i \sim Ga(n, n\lambda)$$

$$\hat{\lambda} \sim \text{inv-}Ga(n, n\lambda)$$